Nakka Chandrakiran

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.58
Sri chaitanya junior college jagtail	Class XII (Telangana Board of Intermediate Education.)	2017	8.20%
Zilla parishad high school yashwanthraopet	Class X (Board of Secondary Education, Telangana state)	2015	83.40%

SCHOLASTIC ACHIEVEMENTS

- **Final-round candidate, Learn in Nippon (LIN) 2026**, Suzuki Innovation Centre × Next Bharat Ventures: cleared multi-stage screening for a fully sponsored India-Japan innovation program *(Dec 2025)*
- **Amazon ML Challenge 2026** -- All-India Rank (Top 400) in entity-matching de-duplication challenge; optimized for macro F0.5 scoring and handling unseen data distributions.

PUBLICATIONS & CONFERENCE PAPERS

Guidance and Navigation Control of Unmanned Surface Vehicle,"*Proceedings of the ASME 2026 International Mechanical Engineering Congress and Exposition (IMECE2026)*, Chennai, India, Sept 2026. [Paper No. IMECE2026-188997] :

- Developed and trained a neural network on 150,000 simulation samples for autonomous waypoint tracking. Designed an 18-input, 2-output model to predict surge force and yaw moment, incorporating heading-error and yaw-rate feedback for closed-loop control. Evaluated tracking performance under simulated wind and wave disturbances.

M.TECH THESIS

AI-Assisted Cross-Domain Communication and Sensor Fusion Architecture for Cooperative USV-ROV Systems:

- Developed a simulated cooperative USV-ROV system in **Gazebo using ROS 2 Jazzy,** in which both the catamaran Unmanned Surface Vehicle (USV) and the BlueROV2 Remotely Operated Vehicle (ROV) operate autonomously. The USV carries [sensors, e.g., GPS, IMU, side-scan sonar] and the ROV carries [sensors, e.g., IMU, depth sensor]. Sensor fusion and cross-domain communication between the two vehicles are [implemented / under development]. Future work will apply AI-based side-scan sonar tracking to localize the ROV at varying relative angles, followed by real-water validation.

PROJECTS

- **Retrieval-Augmented Generation System for USV-ROV Robotics Research:** Developed an end-to-end RAG pipeline for semantic question-answering over marine robotics research papers using LangChain, FAISS, Sentence Transformers, and Groq LLM. Implemented REST APIs using FastAPI with Pydantic validation, error handling, and automated testing using pytest. Containerized the application using Docker and Docker Compose, incorporating automated health checks and CPU/memory resource limits for reproducible deployment.
- **Reinforcement Learning-Based Mobile Robot Navigation:** Trained a Soft Actor-Critic (SAC) policy for 400,000 simulation steps using lidar observations, goal information, and laser-odometry estimates, achieving **95% goal-reaching success across 100 held-out simulation episodes**, with 2% collisions and 3% timeouts. Deployed the policy on a **Jetson Orin Nano** using ROS 2 and Python, integrating YDLIDAR X2, RF2O odometry, EKF, and an ESP32 DDSM motor controller to demonstrate physical robot goal reaching. Implemented RViz goal selection, velocity control, obstacle-based speed blocking, and command timeouts.

- **Neural Network-Based Control of Unmanned Surface Vehicle (USV)** -- Designed and compared PID, Hybrid Fuzzy-PID, and Neural Network controllers for autonomous trajectory tracking, compensating for wind, wave, and current disturbances; evaluated via simulation on a 3-DOF dynamic model
- **Autonomous Navigation of Mobile Robots** -- Developed an autonomous SLAM (Simultaneous Localization and Mapping) pipeline for a mobile robot on ROS2, integrating data from a YDLIDAR X2 (2D LiDAR) and DDSM210 wheel encoder odometry through an Extended Kalman Filter (EKF) to produce accurate real-time pose estimates and generate a map of unknown environments, enabling the robot to autonomously localize itself as it moved.

- **RRR Robotic Manipulator (ROS 2, Gazebo)** -- Developed real-time simulation and physical implementation with FK/IK for synchronized robot motion and end-effector positioning
- **ESP32-Based Flight Controller for F450 Frame** -- Built a custom quadrotor flight controller from scratch using ESP32, MPU6050 IMU, FlySky FS-iA6B receiver, and SimonK ESCs; implemented quad-X motor mixing and flight-tested PID gains for stable flight

SKILLS

- Programming & Frameworks: Python, C++, ROS/ROS 2
- Deep Learning: PyTorch, Keras, Transformers, LLM fine-tuning, VLMs (Vision-Language Models), RAG (Retrieval-Augmented Generation)
- MLOps & APIs: FastAPI, Docker, Docker Compose, REST APIs
- Robotics & Simulation Tools: MATLAB/Simulink, Gazebo, Isaac Sim, SLAM
- Control & Autonomous Systems: PID, Fuzzy Logic, Neural Network-based control,sensor fusion (IMU, GPS, LiDAR, Sonar, Radar), computer vision, autonomous navigation
- Design & Mechanical: CAD (Fusion 360), mechanical design, prototyping

- **Reinforcement Learning:** Soft Actor-Critic (SAC), Proximal Policy Optimization (PPO), Deep Q-Networks (DQN).

- Embedded & Hardware Platforms: Arduino, ESP32, Raspberry Pi, Jetson, Pixhawk

POSITIONS OF RESPONSIBILITY

Teaching Assistant :

- ME2050 Instrumentation
- ME5770 Fundamentals of Robotics
- ME5051 Sensors and Actuators Lab

M.Tech Head and Mentor, Robotix Club:

- Led club operations and mentored junior members on building robots from scratch, covering material selection, electronics/motor sourcing, assembly, and troubleshooting

HOBBIES & EXTRACURRICULAR ACTIVITIES

- **Cooking** -- enjoy experimenting with recipes as a creative outlet outside academics
- **Farming** -- hands-on experience with agricultural work, rooted in a farming background
- **Football** -- active team sport player, regularly play for fitness and teamwork

- **Coordinator, Autonomous Robotics Summer Immersion & Simulation Grand Prix 2026 (IIT Hyderabad × George Mason University)** -- Co-conducted a robotics workshop (BARN Challenge) teaching ROS 2 and Gazebo-based SLAM to participants